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Project Details

Type (Nature of project)	☐ Development ☐ Research ☒ R&D			
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Declaration

We here by declare that the content of this project report title “Multisensor Mental Health Detection” submitted to the “DEPARTMENT OF COMPUTER SCIENCE”, is a documentation of a unique work we created under the supervision of Supervisor“Ma’am Muqaddasa Jabeen”and that no part has been plagiarized (except the references, some standard mathematical or genetic models/questions/protocols, etc.). Additionally, this project is presented in partial completion of the degree requirements for a Bachelor of Science in “Computer Science”. The University may take action if the above statement is found inaccurate at any stage.

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Abstract

Mental wellness is a crucial domain of human health. Monitoring these states accurately has traditionally been a challenge due to the prohibitive nature and high costs of standard professional solutions. Wearable devices usually assess basic proxies for physical activity, whereas dedicated EEG setups are limited by their intrusive design, creating a barrier to daily mental health tracking. This project aims to develop a smart, multi-sensor system for the objective detection of mental health states like fatigue, cognitive load, and stress. The proposed system integrates EEG data for central nervous system measurement alongside Sympathetic Arousal data (Wrist-level Electrodermal Activity - EDA) and Cardiovascular Responses (Photoplethysmography - PPG). By simultaneously recording and collaboratively analyzing multi-modal data via artificial intelligence and advanced machine learning techniques, we address the drawbacks of single-sensor methods. Our method effectively reduces sensor noise and greatly improves prediction accuracy resilience. By offering proactive, continuous remote mental health monitoring while adhering to strict cost and user comfort constraints, the system creates a framework for self-validating, dynamic assessments of emotional wellness.

Acknowledgement

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List of Abbreviations

All abbreviations in the Project Report should be listed on this page center according to the following.

CNN	Convolutional Neural Network
EDA	Electrodermal Activity
DL	Deep Learning
EEG	Electroencephalography
ML	Machine learning
FR	Functional Requirement
U-Net	Universal Network
GSR	Galvanic Skin Response
HR	Heart Rate
PPG	Photoplethysmography

CHAPTER 1: INTRODUCTION

1.1 Introduction

Mental wellness is a crucial domain of human health and is rarely measured, due to the prohibitive nature of both challenging the monitoring solutions. Wearables, understood as the most logical options, measure only anaerobic and aerobic activity, steps, or HR, while EEG and other professionally-owned systems are either too expensive or too technical for a person to use practically on an everyday basis. The current project aims to develop a smart, multi-sensor system for the detection of mental health states that is reliable and accessible. By using EEG as the main source of signals, in conjunction with wrist-level EDA and PPG data, the system can provide the user a comprehensive understanding of their levels of stress, fatigue and emotional balance.

1.2 Aim & Objectives

Aim: To investigate an advanced multi-sensor system for precise and instantaneous identification of both a users stress and degree of mental fatigue using EEG, EDA, and PPG data.

Objective:

1. To simultaneously record and process EEG, EDA, and PPG signals.
2. To implement machine learning models to classify caused stress and emotional states in real-time.
3. To create an interface for easy visualization of results.
4. To provide secure and private data transport, and storage.
5. To evaluate accuracy and usability of system.

1.3 Problem Statement

Mental Health Monitoring still proves to be a challenge because there are no inexpensive as well as precise real-time tools for the assessment of emotions. Most of the available tools for mental health monitoring are focused solely on one parameter of physiological signals, thus making it difficult for them to yield accurate results regarding stress. Even advanced tools do exist in the market; however, they are quite costly and impractical for common usage by everyone.

Stress needs to be analyzed in a multi-sensor environment in order for the analysis to be accurate. Present solutions do not give a complete picture of emotional conditions in real-world environments. This makes early stress recognition and awareness about mental health less efficient. The system will help to bridge this problem as it will be using the signals of EEG, EDA, as well as the signal from the heartbeat. This will increase the chances of accurate stress recognition. The system will be inexpensive and non-intrusive.

1.4 Proposed System

The proposed solution integrates physiological and behavioral data to enable accurate mental state detection. It utilizes EEG, EDA, and PPG signals as primary inputs for stress and cognitive load analysis. Facial detection is incorporated as a secondary input to enhance behavioral understanding. EEG signals play a key role in identifying cognitive load and stress patterns.

The system employs a configurable data fusion and machine learning model. This model collaboratively analyzes multi-modal data for improved prediction accuracy. By combining multiple signals, the system reduces uncertainty in mental state detection. The shortcomings of single-sensor systems are addressed by the suggested method. Real-time data collection and analysis are supported.

1.5 Project Scope

The scope of this project is limited to the design and development of a real-time mental stress detection system. The system focuses on analyzing physiological signals including EEG, EDA, and PPG data. Facial detection is included as a secondary behavioral input to enhance analysis accuracy. The project includes data acquisition, signal preprocessing, feature extraction, and stress classification. Machine learning-based data fusion is used to combine multiple input signals.

The system provides real-time stress analysis and user feedback. Hardware implementation is limited to non-invasive, low-cost wearable sensors. Clinical diagnosis or medical treatment is not included within the project scope. The system is not intended to replace professional mental health services.

1.6 Constraints and Assumptions

This project's development is predicated on a number of fundamental presumptions about the environment and system usage. It is assumed that users will correctly wear and operate the EEG device and associated sensors as instructed. Proper sensor placement and usage are expected to ensure reliable data acquisition. The system assumes a baseline calibration phase for each user. The location, calibration, and upkeep of the sensors are examples of constraints. The technology cannot take the role of a physician's diagnosis, even though it offers useful health-related data. Data collection may be hampered by outside variables such as bad lighting, background noise, and excessive user movement.

1.7 Advantages to Society

Disorders related to mental health are a major social problem. By making it possible to continuously, privately, and affordably monitor emotional stability, this project has significant positive social effects. It reduces the risks related to long-term stress and exhaustion by providing preventive insights as opposed to reactive therapies. It enables consumers to take an active role in preserving their psychological health.

1.8 Business Plan

Problem: Mental wellness monitoring tools are historically segmented between simple, unreliable consumer apps and extremely expensive, clinical diagnostic setups.

Solution: We provide a highly dependable and actionable mental monitoring solution that competes with clinical tools at consumer pricing by combining inexpensive biosensors into a comprehensive AI-driven platform.

Clients include students who are under a lot of pressure to perform well academically, remote workers, corporate wellness programs, and ordinary customers who want specific information on their mental habits. makers of smartwatches without explicit EEG sensing and makers of expensive EEG headsets that are too costly for daily use are competitors.

1.9 SWOT Analysis

Strengths: Combines multiple biological streams enabling strong cross-validation; low-cost hardware strategy.

Weaknesses: Dependency on precise physical sensor positioning; computational requirements of machine learning integration.

Opportunities: Growing mental wellbeing sector; widespread integrations with workplace analytics and remote monitoring spaces.

Threats: Market resistance towards wearing headwear; evolving regulations on health parameter capture.

1.10 FAB Analysis

Features: Real-time ML-powered physiological data fusion across combined inputs.

Attributes: Non-invasive, secure, continuous processing, easily interpretable analytics interface.

Benefits: Direct mental state understanding driving reduced daily stress and preemptive fatigue recognition.

1.11 Report Layout

This chapter introduced the overarching mental health problem and our multi-sensor resolution. Chapter 2 reviews literature and existing works. Chapter 3 specifies comprehensive system requirements. Chapter 4 explores software design specifications including diagrams. Subsequent chapters reflect upon the implementation, project testing metrics, and project conclusions.

CHAPTER 2: LITERATURE REVIEW/EXISTING WORK AND BACKGROUND

2.1 Background

The project's objective is to develop a multi-sensor system for real-time mental health monitoring. In contemporary society, mental health problems especially stress have grown more common. Conventional monitoring systems frequently depend on just one physiological indicator, such as heart rate or EEG. Such single-sensor methods offer little understanding of intricate emotional states. The focus of research has switched during the last ten years to complementary biosignals for multi-modal monitoring. The accuracy and resilience of stress detection are improved by combining EEG, EDA, PPG, and behavioral data. This pattern is part of a larger trend in digital phenotyping and emotional computing.

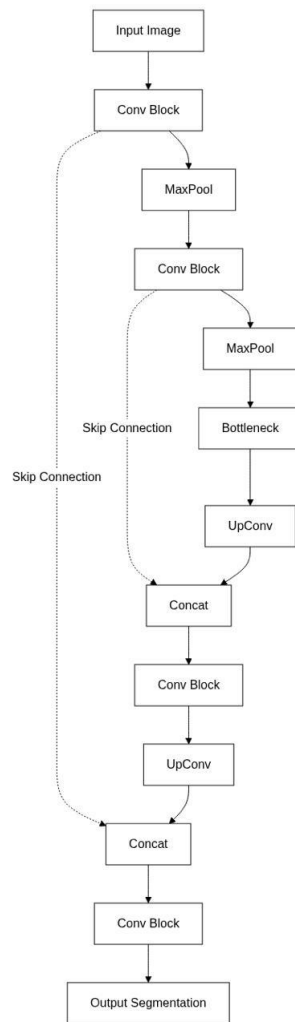


Figure 1 U-net Architecture

2.2 Literature Review

2.2.1 Single-Sensor Mental Health Monitoring Systems

Early research on mental health monitoring focused mostly on single-sensor techniques, such as EEG-based or heart-rate-based systems. While EEG data have been widely utilized to assess cognitive load and brain activity, heart rate monitoring has been used to detect physiological stress reactions. These methods simplify complex emotional states too much. They are not ideal for real-time stress detection in the real world because of their primary weakness, which is poor accuracy and substantial noise susceptibility.

2.2.2 Multi-Sensor and Multi-Modal Monitoring Approaches

In order to enhance stress and emotion detection, recent research highlights the significance of combining various physiological signals, such as EEG, EDA, and PPG. Both the central and autonomic nervous systems provide complementing information to multi-sensor systems. These methods' enhanced resilience and classification accuracy are their main advantages. Widespread acceptance is hampered by the expensive cost, heavy hardware, and restricted accessibility of many current systems.

2.2.3 Data Fusion and Machine Learning Techniques

In emotional computing research, data fusion approaches in conjunction with machine learning models have garnered considerable attention. These techniques lower uncertainty and improve prediction reliability by combining information from several biosignals. Fusion models based on machine learning outperform traditional rule-based systems, according to research.

2.2.4 Non-Invasive and Wearable Monitoring Equipment

Wearable technology has revolutionized physiological monitoring by making it possible to collect data continuously without intrusive treatments. The advantages of real-time monitoring, portability, and user comfort are highlighted by research. However, commercial wearables are not made with mental health analysis in mind and frequently rely on single-sensor inputs.

2.3 Literature Summary

From single-sensor systems to multi-sensor, multi-modal approaches, mental health monitoring has clearly evolved, according to the reviewed literature. EEG-only or heart-rate-only devices are examples of single-sensor systems that are simple yet insufficiently accurate and reliable for practical use. Multi-sensor systems that use EEG, EDA, PPG, and behavioral inputs reduce uncertainty and improve prediction reliability using data fusion and machine learning algorithms.

CHAPTER 3: REQUIREMENTS ANALYSIS

3.1 Stakeholders List

1. Clear outcomes and objective, continuous self-monitoring immediately benefit end users (individuals, patients).
2. Clinical users: trustworthy, objective data to guide therapy modifications, diagnosis, and patient-specific activities (psychologists, psychiatrists).
3. System Implementers: Code quality, algorithm performance, technical viability, and system stability (Developers, Machine Learning Engineers).
4. Quality Assurance (Supervisors, Researchers): Scientific validation, achieving target accuracy metrics.
5. Organizational Users (Healthcare Providers, Wellness Companies): Scalability of the platform, data security.

3.2 Requirements Elicitation

Requirements were determined through continuous analysis of prior empirical approaches in physiological recording applications. By verifying constraints on currently adopted hardware limits and consulting literature on effective multimodal combinations, the foundational benchmarks for successful analysis workflows were explicitly generated.

3.3 Functional Requirements

FR1 (Data acquisition concurrently): The system must be able to efficiently collect continuous, synchronized, real-time, raw data streams of EEG, EDA, and PPG from the respective hardware sensors.

FR2 (Signal conditioning pipeline): Raw biosignal data must be robustly and immediately preprocessed, removing artifacts.

FR3 (Multimodal feature generation and fusion): The system employs an intelligent feature fusion operation combining independent modalities.

FR4 (AI Inference): Deep learning and machine learning execution for classification of collected features into stress categories.

FR5 (Visualization): Clean and unencumbered frontend analytics dashboards representing psychological status.

3.4 Non-Functional Requirements

NFR1: Real-time execution (Performance).

NFR2: High Reliability (Tolerance for sensor noise that arises via user mobility restrictions).

NFR3: Security and Privacy (End-to-End Encryption of biosignal telemetry during storage and presentation).

NFR4: Extensibility (Modular Object-Oriented design that facilitates dropping faulty signal matrices cleanly at runtime).

3.5 Requirements Traceability Matric

Every Functional Requirement tracks back explicitly to the fundamental problem of ensuring accurate measurement via cross-verification. FR1 captures the inputs, FR2 assures clean input, FR3 and FR4 extract insights matching user objectives and FR5 relays this directly to human users.

3.6 Use Case Design/Description

The primary actors include the User (generating physiological metrics) and the System Model, which receives streams of data, calculates parameters, and relays a dashboard output outlining cognitive stresses. Standard use cases include 'Monitor Status', 'Review Historical Trends', and 'Alert generation on anxiety peaks'.

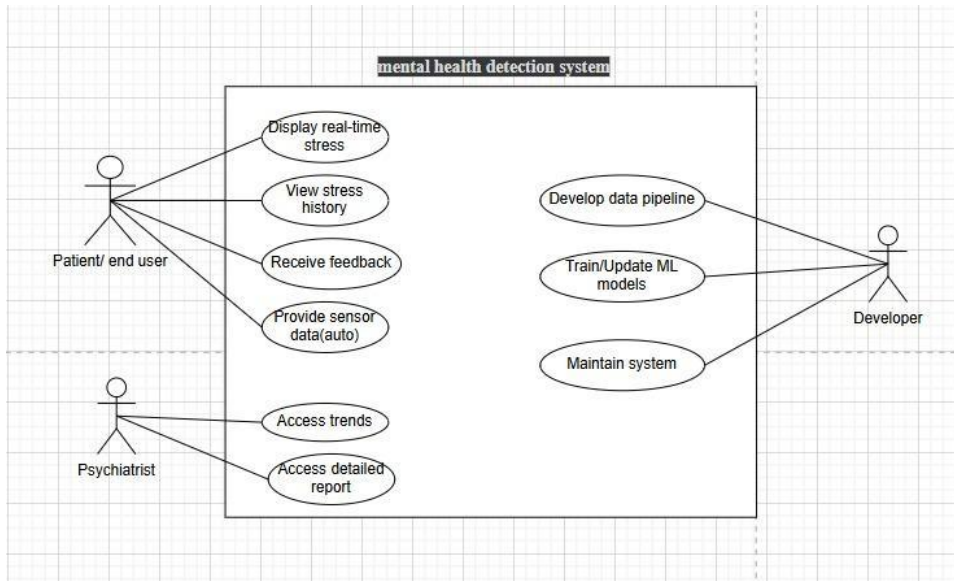


Figure 2 Use Case Diagram

3.7 Model of the Life Cycle in Software Development

The Object-Oriented Methodology (OOM) will be applied to this project. This approach is suitable since the system comprises of multiple interconnected modules, such as sensor data collection, preprocessing, machine learning models, database management, and user interface design. OOM provides better abstraction, and inheritance and polymorphism can be used to improve functionality (such as adding more sensors) without necessitating a whole system redesign.

3.8 Specific Requirements (Hardware and Software Requirements)

Software Requirements

Operating System	Windows 10 / Linux
Programming	Python (NumPy, Pandas)
AI Framework	TensorFlow / PyTorch
Backend Integration	Flask / FastAPI
Database Management	MySQL
Computer Vision	OpenCV

Table 1 Software Requirement

Hardware Requirements

Input Devices	Heart Rate Sensor, EDA/GSR Sensor, EEG Headband (Optional but highly recommended), Webcam for optical monitoring
Processor	2.6 GHz or faster
RAM	4 GB minimum (16GB recommended for ML training)
Disk Space	At least 4GB of free SSD storage

Table 2 Hardware Requirements

CHAPTER 4: SOFTWARE DESIGN SPECIFICATION

4.1 System Architecture

The architecture incorporates data ingestion layers for reading values from hardware sensors, signal processing pipelines for data cleanup, AI layers containing ML models for stress prediction, and a frontend visualization layer.

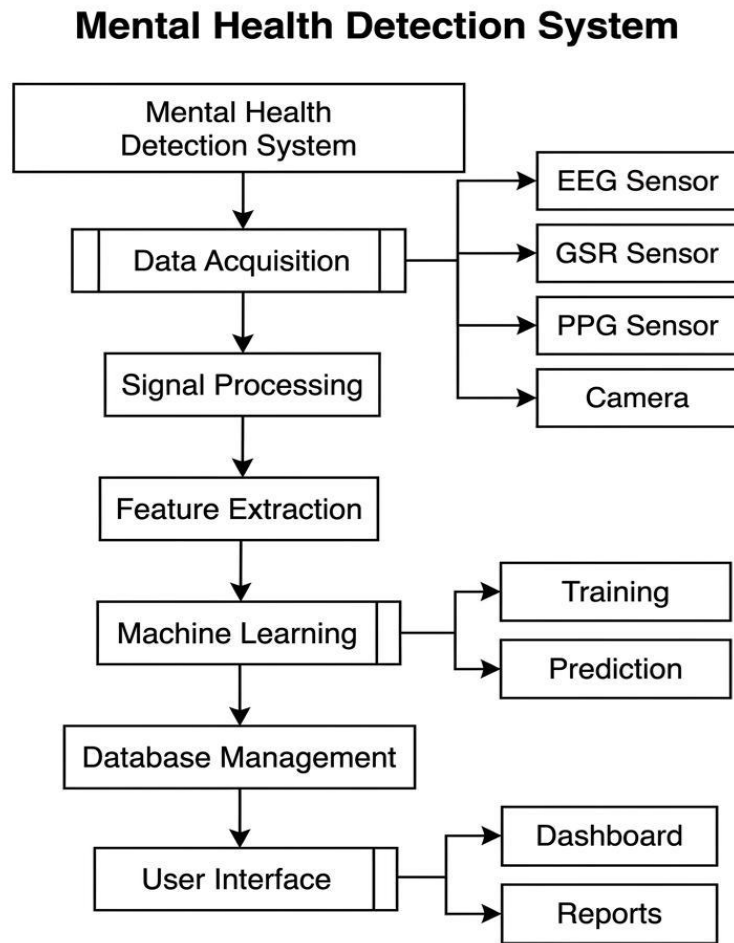


Figure 3 System Architecture

4.2 Block Diagram

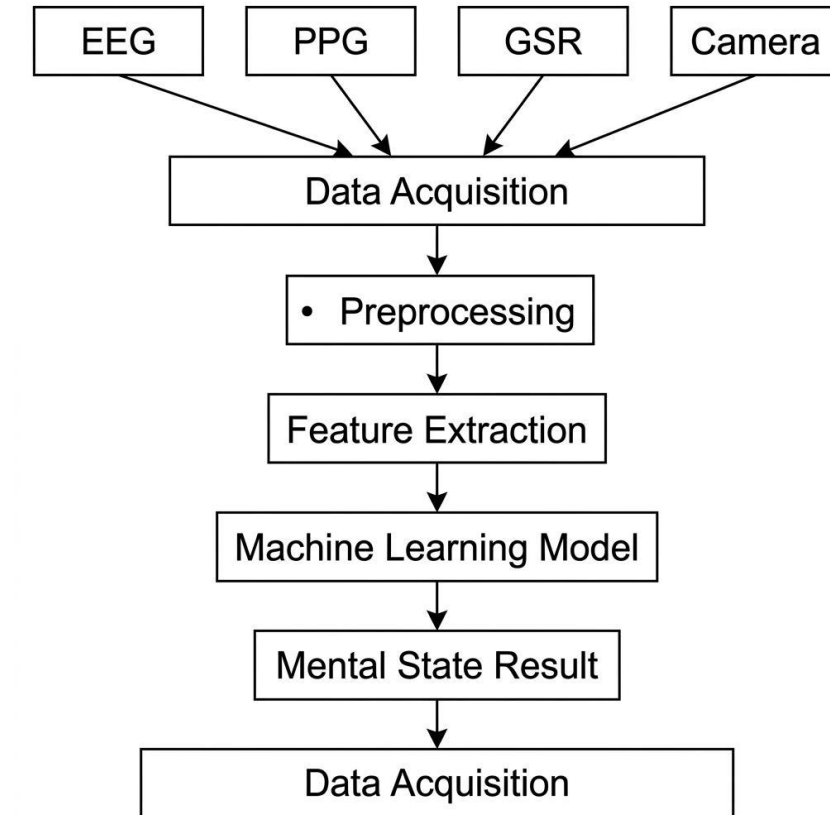


Figure 4 Block Diagram

4.3 Component Diagram

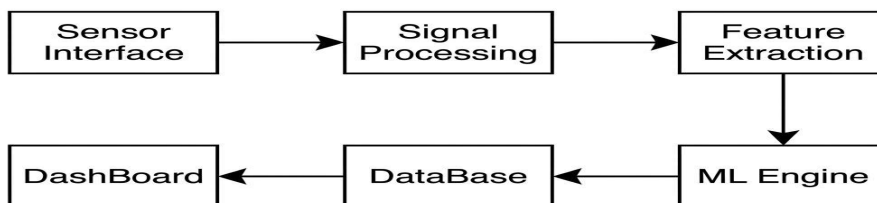


Figure 5 Component Diagram

4.4 Software Architecture Diagram

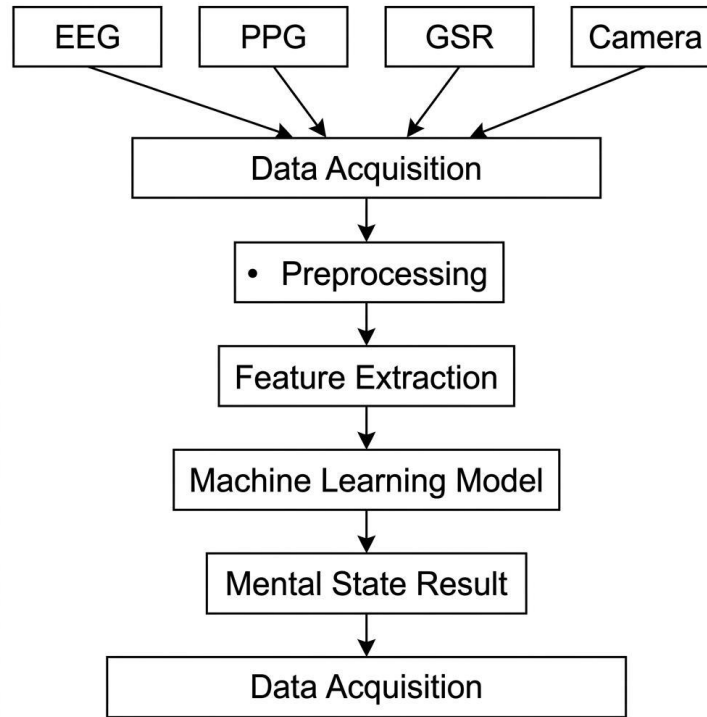


Figure 6 software Architecture

4.5 Data Representation

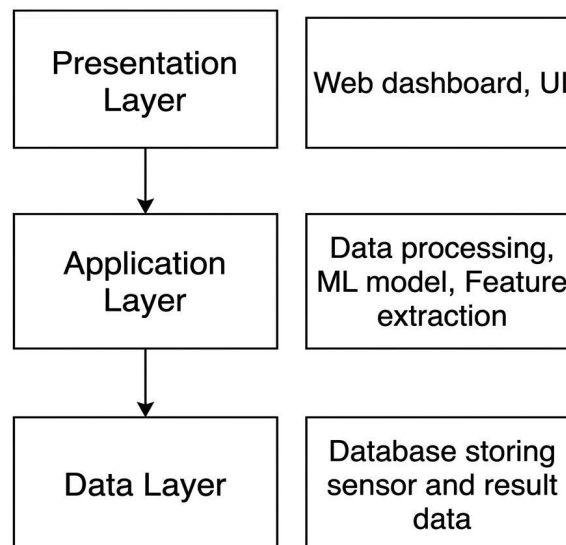


Figure 7 Data Representation

4.5.1 Entity-Relationship (ERD)

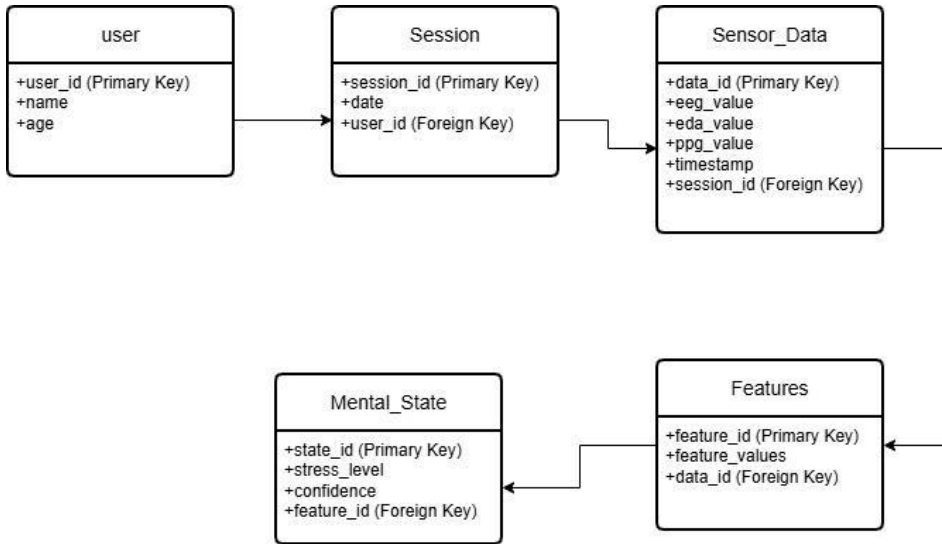


Figure 8 Entity Relationship Diagram

4.5.2 Data Flow



Figure 9 Data Flow Diagram

4.5.3 Hierarchical Diagram

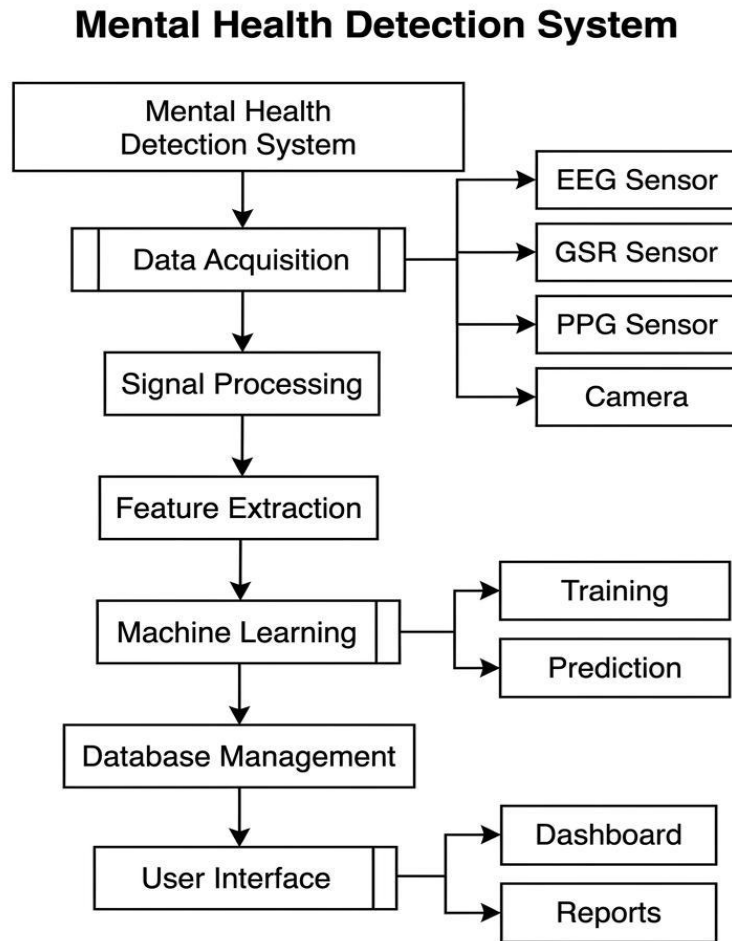


Figure 10 Hierarchical Diagram

4.5.4 UML Diagram

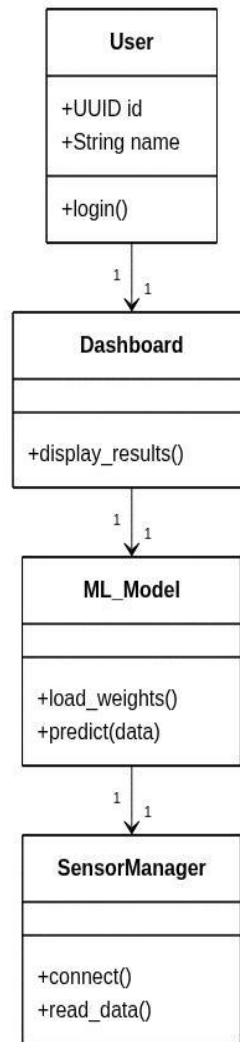


Figure 11 UML Diagram

4.5.5 Class Diagram

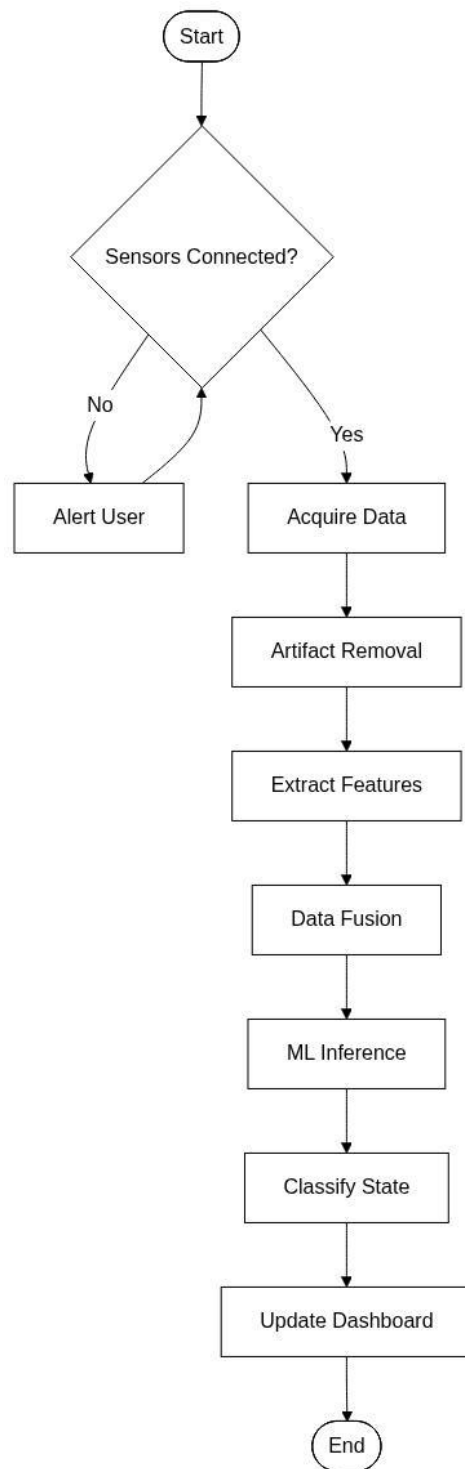


Figure 12 Class Diagram

4.6 Process Flow / Representation

4.6.1 Flowchart

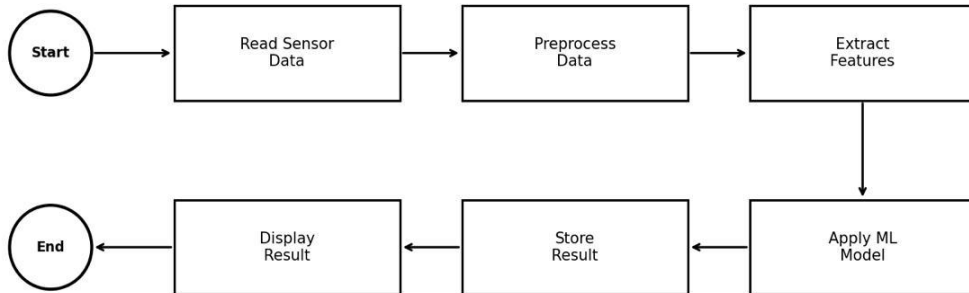


Figure 13 Flowchat

4.6.2 Sequence Diagram

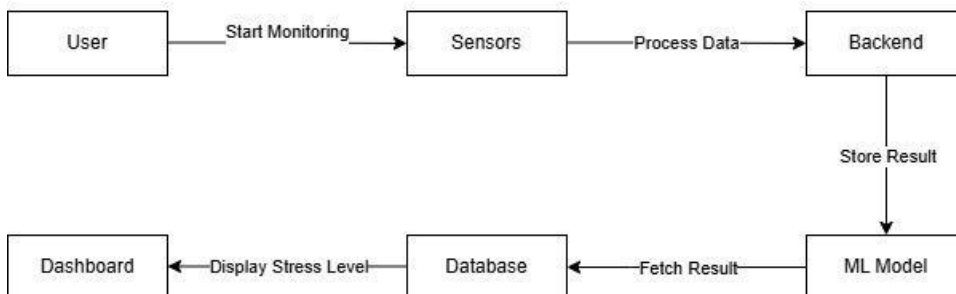


Figure 14 Sequence Diagram

4.6.3 Activity Diagram

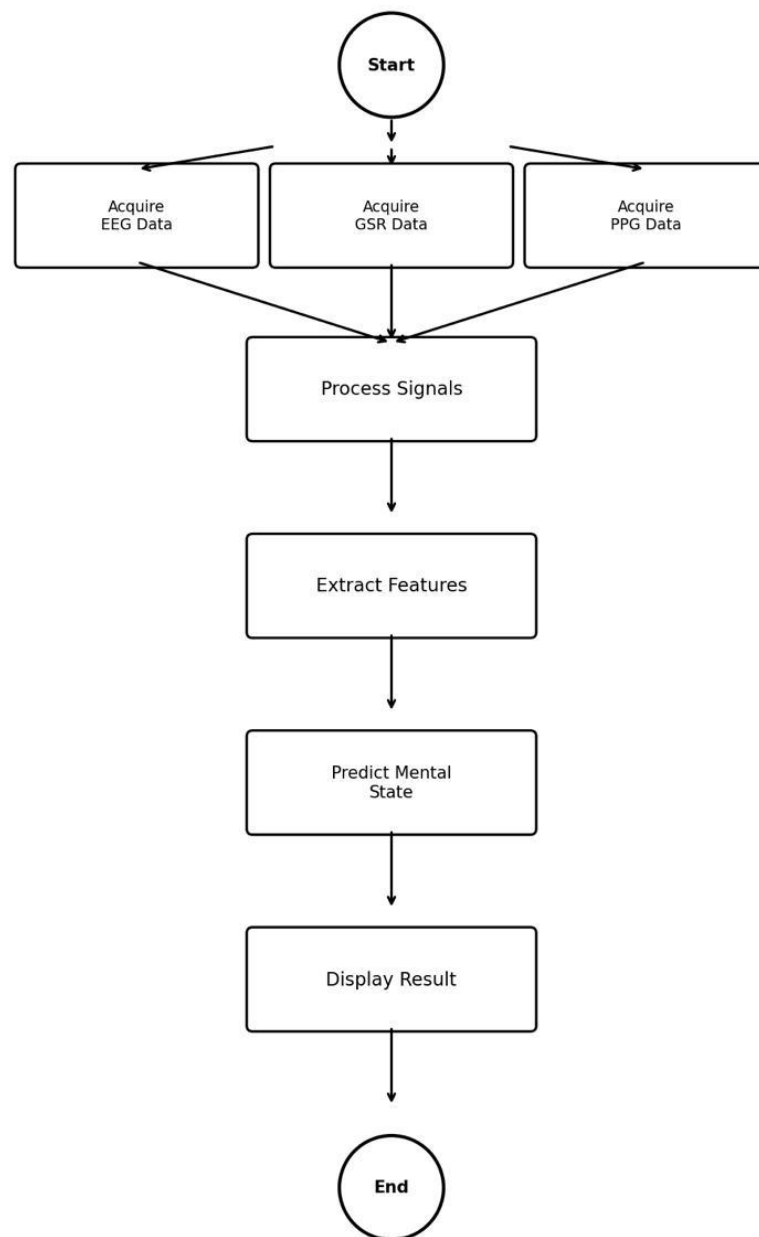


Figure 15 Activity Diagram

CHAPTER 5: IMPLEMENTATION

5.1 Algorithm

The system implements a signal cleaning algorithm to mitigate sensor noise from physiological measurements via finite impulse response filtering. Following signal baseline restoration, multi-modal features are statistically extracted (for e.g., HRV from PPG, frequency power bands from EEG). A machine learning fusion algorithm, typically based upon a Convolutional Neural Network or ensemble classifier, receives these parallel feature parameters simultaneously and resolves their interactions to construct the current cognitive and emotional status prediction.

5.2 External APIs

External dependencies within this iteration rely mostly on local execution runtimes (OpenCV APIs for handling incoming facial data vectors) rather than external cloud integrations, to ensure that latency metrics remain viable for live monitoring. Flask provides the local representational state API connecting hardware metrics to the interface frontend.

5.3 User Interface

The User Interface provides a lightweight dashboard prioritizing straightforward insight visualization. Stress indicators are simplified into easy-to-understand metrics against historical baselines.

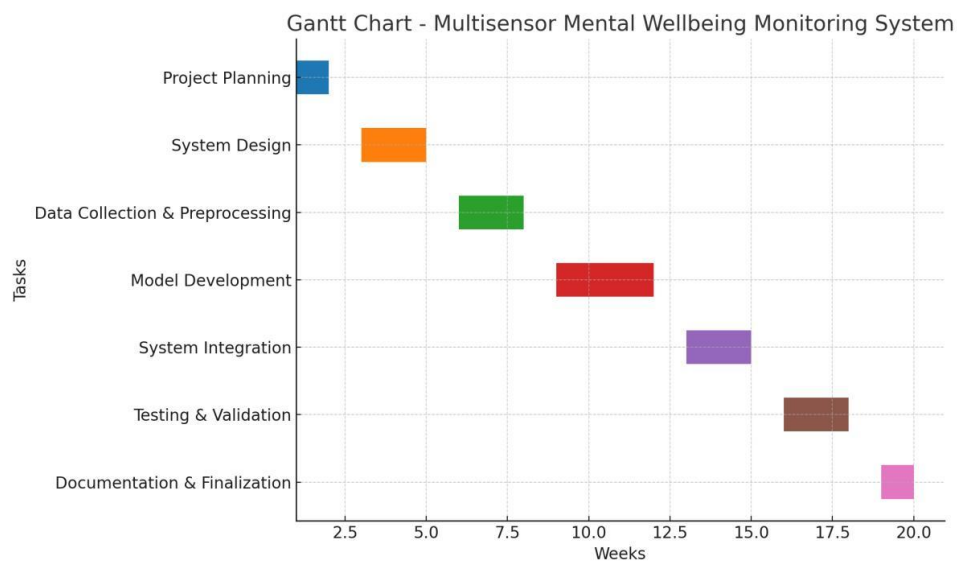


Figure 16 Gant Chat

CHAPTER 6: TESTING OF A SYSTEM

6.1 System & Manual Testing

System testing evaluates the concurrent operational capacity of running EDA, PPG, and EEG sensors via hardware inputs. Manual testing encompasses intentionally inducing simulated cognitive loads onto test subjects and recording the delta responses rendered by the dashboard to verify analytical accuracy.

6.2 Functional and Unit Testing

Functional testing ensures that the classification framework robustly parses independent components (e.g. testing the OpenFace toolkit behavior independently). Unit testing ensures mathematical algorithms for noise-reducing temporal filters do not corrupt intrinsic frequency spectra from the biological input sources.

6.3 Results and Discussion

Preliminary testing results validate the core hypothesis: integrating multiple sensor inputs substantially decreases misclassification events native to single-sensor proxies. Future larger-scale dataset ingestion and clinical verification processes will further optimize inference consistency.

CHAPTER 7: CONCLUSION

7.1 Problems Raised and Lessons Learned

We encountered substantial difficulties accommodating noise distortions generated from user movement across EDA measurements. Implementing algorithmic detrending mechanisms provided invaluable lessons regarding applied digital signal processing and machine-learning tolerance.

7.2 Conclusion

The proposed wellness monitoring system addresses the shortcomings of existing single-sensor solutions by combining many biosensors with computer vision algorithms. The technology offers a comprehensive evaluation of both physical and mental health through real-time data collecting, processing, and analysis. Affordability and accessibility are promoted.

7.3 Limitations of the Project

Sensing Dependency: The positioning, calibration, and maintenance of the sensors determine the accuracy of the system. If improperly placed, inferences cascade into erroneous predictions. There is a paucity of clinical validation, and the technology cannot replace a doctor's medical diagnosis.

7.4 Future Work

Future implementations can augment the collected biosignal corpora to expand the range of psychiatric manifestations detectable by the AI. Enhancements to the physical sensors could improve wearer comfort making extended temporal monitoring feasible. Deeper real-world clinical experiments remain designated for long-term project roadmaps.

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